

Computational Landscape Geometry and Representation (CLG-R v4.0.5): Rigorous Structural Results and Open Dynamical Problems for Continuous 3-SAT Relaxations

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Abstract

Continuous relaxations of Boolean Satisfiability (3-SAT) are widely utilized in neural combinatorial optimization, semidefinite programming, and physics-inspired dynamical solvers. However, the fundamental mathematical mechanism determining why certain continuous representations succeed while other Boolean-equivalent formulations fail has remained unformalized. In this paper, we establish the rigorous analytical foundations of **Computational Landscape Geometry and Representation (CLG-R)** on the compact hypercube $\mathcal{X} = [-1, 1]^N$.

We prove: (1) The Quadratic Hinge relaxation unconditionally generates a central fractional plateau $\mathcal{U}_N = (-1/3, 1/3)^N$ with identically vanishing gradient $\nabla \Phi_{\text{quad}} \equiv \mathbf{0}$, manifesting the exact interior slack (0.5) of the canonical Linear Programming (LP) relaxation; (2) Under structural non-degeneracy (Walsh-Fourier variance $\text{Var}(E_{\text{disc}}) > 0$), the singular sets of Multilinear and Softplus relaxations have Lebesgue measure zero ($\mu(\mathcal{C}_0) = 0$); (3) The Multilinear relaxation is harmonic ($\Delta \Phi_{\text{mult}} \equiv 0$), precluding any interior local minimum via the Strong Minimum Principle, while all interior critical points admit arbitrary-neighborhood lower-energy points via Milnor’s Curve Selection Lemma; (4) Without boundary non-degeneracy hypotheses, every local minimum of Φ_{mult} on a face $\mathcal{F}$ satisfies $\Phi_{\text{mult}}(x^*) = E_{\text{disc}}(v)$ for all $v \in \mathcal{V}(\mathcal{F})$, and every isolated asymptotically stable equilibrium of the projected gradient flow is strictly a discrete vertex in $\{-1, +1\}^N$; (5) The Softplus relaxation factors as $\nabla^2 \Phi_{\text{soft}} = V^T W(x) V \succeq 0$, attaining global strict convexity when $\text{rank}(V) = N$, with condition number bounded by $\kappa(\nabla^2 \Phi) \leq \kappa(W(x))\kappa(V^T V)$; (6) The Lipschitz constant of the gradient field scales as $L_\beta = \Theta(\beta)$, while finite-precision evaluation can suppress exponentially small Softplus gradients into subnormal/underflow regimes, numerically approaching Hinge-like flat behavior; (7) We establish the Universal Centripetal Contraction Theorem for Hinge ($\langle -\nabla \Phi_{\text{quad}}, x \rangle < 0$), proving the absence of stationary points outside the LP polytope Z and universal approach to the LP polytope ($\lim_{t \rightarrow \infty} \text{dist}(x(t), Z) = 0$); (8) For the independent-clause random 3-SAT ensemble, we prove analytically that the normalized volume of the LP polytope satisfies the finite-dimensional lower bound $\mathbb{E}[\mu_{\text{norm}}(Z)] \geq (5/6)^{\lfloor \alpha N \rfloor} \geq e^{-N\alpha \ln(6/5)} > 0$ via the Irwin-Hall distribution of order 3; (9) We analyze dynamical behavior in monotone Horn families: purely negative clauses have competitive off-diagonal sign ($J_{ij} \leq 0$), while general Horn clauses can have mixed Jacobian signs; consequently standard orthant-monotonicity results do not settle the general case, and Hinge separation on Horn DAGs remains open after the singleton collapse under positive unit facts; (10) An adversarial audit refutes the previously claimed complete subcritical separation: an isolated six-clause linear hypertree is the minimal linear-hypertree obstruction with an open basin of positive multilinear residual energy. In the uniform distinct-clause ensemble, $\liminf_{N \rightarrow \infty} \bar{\rho}_{N, \text{mult}}(\alpha) \geq (729/2^{59})\alpha^5 e^{-39\alpha} > 0$ for every fixed

$\alpha > 0$, and $\mathbb{P}_{F, x_0}[\rho_{N, \text{mult}}(F, x_0) \geq (729/2^{60})\alpha^5 e^{-39\alpha}] \rightarrow 1$. Pointwise convergence of the projected multilinear flow remains valid through nonsmooth Kurdyka–Łojasiewicz theory, but convergence to zero residual energy does not. For the random 3-SAT ensemble, the remaining clustering-regime conjecture is therefore comparative ($\rho_{\text{quad}} > \rho_{\text{mult}}$), not positivity of the multilinear residual.

Keywords: Computational Complexity, Differential Topology, Morse Theory, Continuous Relaxations, MAX-3-SAT, Dynamical Systems, Convex Optimization, Spin Glasses.

1 Introduction and Theoretical Framework

The Cook-Levin Theorem [6, 13, 17] established 3-SAT as the canonical NP-complete problem. Over decades, numerous continuous embeddings have been proposed to solve or approximate SAT using continuous dynamical systems [8, 12, 16, 18, 21].

The central thesis of the **Computational Landscape Geometry and Representation (CLG-R)** framework is that *Boolean equivalence does not imply dynamic or algorithmic equivalence in continuous space*. Two potential functions $\Phi_1, \Phi_2 : [-1, 1]^N \rightarrow \mathbb{R}$ that agree perfectly on discrete vertices $\{-1, 1\}^N$ can induce fundamentally incompatible critical set topologies, Morse indices, and gradient flow reachabilities.

1.1 The Three Canonical Continuous Relaxations and Physical Identifications

Let F be a 3-CNF formula with M clauses over N Boolean variables $s \in \{-1, +1\}^N$. For each clause $c \in \{1, \dots, M\}$, let $\sigma^{(c)} \in \{-1, 0, +1\}^N$ denote its literal polarity vector, where $\sigma_j^{(c)} = +1$ if variable j appears positive, $\sigma_j^{(c)} = -1$ if negated, and 0 if absent. The clause violation affine function is:

$$g_c(x) = -\frac{1}{2} \left(1 + \sum_{j \in c} \sigma_j^{(c)} x_j \right) = -\frac{1}{2} (1 + \sigma^{(c)} \cdot x). \quad (1)$$

At discrete vertices $s \in \{-1, +1\}^N$, $g_c(s) = 1$ if clause c is violated, and $g_c(s) \leq 0$ if satisfied. The three canonical relaxations on the hypercube $\mathcal{X} = [-1, 1]^N$ are:

1. Quadratic Hinge Relaxation:

$$\Phi_{\text{quad}}(x) = \sum_{c=1}^M [\max(0, g_c(x))]^2. \quad (2)$$

LP Identification: The zero-level set $Z = \{x \in \mathcal{X} \mid g_c(x) \leq 0, \forall c\}$ is the canonical clause-wise LP relaxation. Under $z_j = (1 + x_j)/2$, a clause constraint is $\sum_{j: \sigma_j^{(c)} = +1} z_j + \sum_{j: \sigma_j^{(c)} = -1} (1 - z_j) \geq 1$ [11].

2. Multilinear Harmonics Relaxation:

$$\Phi_{\text{mult}}(x) = \sum_{c=1}^M \prod_{j \in c} \frac{1 - \sigma_j^{(c)} x_j}{2}. \quad (3)$$

Physical Identification: $\Phi_{\text{mult}}(x)$ is rigorously the **naive mean-field Hamiltonian** of statistical physics [4, 23, 26]:

$$\Phi_{\text{mult}}(x) = \mathbb{E}_{s \sim \prod_{i=1}^N \text{Bern}\left(\frac{1+x_i}{2}\right)} [E_{\text{disc}}(s)]. \quad (4)$$

For 3-XOR-SAT, Φ_{mult} matches the diluted p -spin spin glass Hamiltonian [10, 19, 25]:

$$\Phi_{\text{mult}}(x) = \frac{1}{2} \sum_e (1 - J_e x_i x_j x_k). \quad (5)$$

3. Softplus Convex Relaxation:

$$\Phi_{\text{soft}}(x) = \sum_{c=1}^M \frac{1}{\beta} \ln \left(1 + e^{\beta g_c(x)} \right), \quad \beta > 0. \quad (6)$$

1.2 Dynamical Flow and Structural Hypotheses

The governing continuous dynamical system is the **projected gradient flow**:

$$\dot{x}(t) = \Pi_{T_{\mathcal{X}}(x(t))} (-\nabla \Phi(x(t))) \quad (7)$$

where $T_{\mathcal{X}}(x)$ is the tangent cone to $\mathcal{X}$ at x , and Π_K is orthogonal projection onto the closed convex cone K [22]. The temporal discretization is pure projected Euler: $x^{(t+1)} = \text{clip}(x^{(t)} - \eta \nabla \Phi(x^{(t)}), -1, 1)$, without artificial box penalties.

The foundational structural hypotheses are:

- **(H1 - Irreducibility)**: Each clause contains exactly three distinct variables ($|\{i, j, k\}| = 3$), without trivial tautologies.
- **(H2 - Variable Connectivity)**: Every variable appears in at least one clause ($\deg(x_i) \geq 1$).
- **(H3' - Walsh-Fourier Non-Degeneracy)**: The discrete energy $E_{\text{disc}} : \{-1, 1\}^N \rightarrow \mathbb{Z}_{\geq 0}$ is non-constant: $\text{Var}_{s \sim \text{Unif}(\mathcal{V})}(E_{\text{disc}}(s)) > 0$. By Parseval's identity on the Boolean hypercube [23]:

$$\sum_{S \subseteq \{1, \dots, N\}, S \neq \emptyset} \hat{E}_{\text{disc}}(S)^2 = \text{Var}(E_{\text{disc}}) > 0, \quad (8)$$

guaranteeing that $\Phi_{\text{mult}} \not\equiv \text{const.}$

1.3 The Four Structural Levels of Critical Geometry

We formalize the landscape hierarchy across four structural levels:

1. Level 1 (Static Critical Geometry):

$$\mathcal{C}_{\text{spur}}(\Phi) \equiv \{x \in \mathcal{X} : \nabla \Phi(x) = \mathbf{0}, E_{\text{disc}}(\text{sign}(x)) > 0\}.$$

2. Level 2 (Local Stability):

$$\mathcal{S}_{\text{spur}}(\Phi) \equiv \{x^* \in \mathcal{X} : x^* \text{ is Lyapunov-stable for } \mathcal{D}_{\text{proj}}, E_{\text{disc}}(\text{sign}(x^*)) > 0\}.$$

3. Level 3 (Attraction Basins):

$$\mathcal{B}_{\text{spur}}(\Phi, \mathcal{D}) \equiv \{x_0 \in \mathcal{X} : \omega(x_0; \mathcal{D}) \subseteq \{x \in \mathcal{X} : E_{\text{disc}}(\text{sign}(x)) > 0\}\}.$$

4. Level 4 (Spurious Dynamical Mass): $\mathcal{M}_{\text{spur}}(\Phi, \mathcal{D}) \equiv \mu_{\text{norm}}(\mathcal{B}_{\text{spur}}(\Phi, \mathcal{D}))$, with μ_{norm} the normalized Lebesgue measure on $\mathcal{X}$.

5. Level 5 (Residual Violation Density): For a finite formula F with M clauses, initial condition x_0 , and flow state $x_{F, x_0}(T)$, define the random finite-size residual density

$$\rho_{N, T}(F, x_0) \equiv \frac{1}{M} E_{\text{disc}}(\text{sign}(x_{F, x_0}(T))).$$

When the projected flow converges, write

$$\rho_N(F, x_0) \equiv \lim_{T \rightarrow \infty} \rho_{N, T}(F, x_0).$$

The ensemble-and-initialization expectation is denoted

$$\bar{\rho}_N(\alpha) \equiv \mathbb{E}_{F, x_0}[\rho_N(F, x_0)].$$

Thus probability statements refer to the random variable $\rho_N(F, x_0)$, while expectation statements refer to $\bar{\rho}_N(\alpha)$. This convention removes the earlier ambiguity in which the same symbol $\rho(\alpha)$ was used both for a random residual density and for its expectation.

Throughout, fix one deterministic tie convention for $\text{sign}(0)$ (for definiteness, $\text{sign}(0) = +1$); whenever a result is tie-independent, this is stated explicitly.

We also distinguish the random-formula models: $\mathcal{E}_{\text{iid}}(N, \alpha)$ denotes $M = \lfloor \alpha N \rfloor$ independent signed-clause draws, while $\mathcal{E}_{\text{unif}}(N, \alpha)$ denotes the uniform M -subset of the $Q_N = 8\binom{N}{3}$ distinct signed clauses.

2 Theorem 1: Central Fractional Box and Linear Relaxation Slack

Lemma 2.1 (Simultaneous Clause Inactivity). *If $g_c(x) \leq 0$ for all $c \in \{1, \dots, M\}$, then $\Phi_{\text{quad}}(x) \equiv 0$ and $\nabla \Phi_{\text{quad}}(x) \equiv \mathbf{0}$.*

Theorem 2.2 (The Central Fractional Box Theorem). *For any 3-CNF formula F satisfying (H1) and (H2), the zero-gradient set $Z(\nabla \Phi_{\text{quad}}) \equiv \{x \in \text{int}(\mathcal{X}) \mid \nabla \Phi_{\text{quad}}(x) = \mathbf{0}\}$ contains the open central hypercube $\mathcal{U}_N = (-1/3, 1/3)^N$. Consequently:*

1. $\Phi_{\text{quad}}(x) \equiv 0$ and $\nabla \Phi_{\text{quad}}(x) \equiv \mathbf{0}$ on all of $\mathcal{U}_N$.
2. The normalized volume of the LP polytope satisfies $\mu_{\text{norm}}(Z) \geq \mu_{\text{norm}}(\mathcal{U}_N) = (1/3)^N$.
3. Under (H3'), the spurious critical set $\mathcal{C}_{\text{spur}}(\Phi_{\text{quad}})$ contains all clause-violating orthants within $\mathcal{U}_N$, satisfying:

$$\mu_{\text{norm}}(\mathcal{C}_{\text{spur}}(\Phi_{\text{quad}})) \geq \left(\frac{1}{6}\right)^N > 0. \quad (9)$$

For unsatisfiable (UNSAT) formulas, $\mu_{\text{norm}}(\mathcal{C}_{\text{spur}}(\Phi_{\text{quad}})) \geq \mu_{\text{norm}}(Z) \geq (1/3)^N$.

Proof. Let $x \in \mathcal{U}_N$. Then $|x_i| < 1/3$ for all $i \in \{1, \dots, N\}$. For any clause $c = (\ell_1 \vee \ell_2 \vee \ell_3)$:

$$\sum_{j \in c} \sigma_j^{(c)} x_j \geq - \sum_{j \in c} |\sigma_j^{(c)} x_j| = - \sum_{j \in c} |x_j| > -3 \times \frac{1}{3} = -1. \quad (10)$$

Substituting into $g_c(x)$:

$$g_c(x) = -\frac{1}{2} \left(1 + \sum_{j \in c} \sigma_j^{(c)} x_j \right) < -\frac{1}{2}(1 - 1) = 0. \quad (11)$$

Thus $g_c(x) < 0$ holds simultaneously for all M clauses. By Lemma 2.1, $\Phi_{\text{quad}}(x) \equiv 0$ and $\nabla \Phi_{\text{quad}}(x) \equiv \mathbf{0}$ on all of $\mathcal{U}_N$. Under (H3'), choose any Boolean assignment $s \in \{-1, +1\}^N$ that violates at least one clause. The full sign orthant $\{x \in \mathcal{U}_N : \text{sign}(x) = s\}$ has normalized measure $(1/6)^N$ and is contained in $\mathcal{C}_{\text{spur}}(\Phi_{\text{quad}})$. For UNSAT formulas, every full sign orthant in $\mathcal{U}_N$ is spurious, so the whole central box contributes. $\square$

Remark 2.3 (Geometric Manifestation of LP Relaxation Slack). *For a fixed clause define its literal-satisfaction coordinates $q_{cj} = (1 + \sigma_j^{(c)} x_j)/2 \in [0, 1]$. Then $g_c(x) \leq 0 \iff \sum_{j \in c} q_{cj} \geq 1$. At the origin every $q_{cj} = 1/2$, so every 3-literal clause has left-hand side $3/2$ and exact **fractional slack** of 0.5. The plateau $\mathcal{U}_N$ provides the geometric realization of this fractional interior slack.*

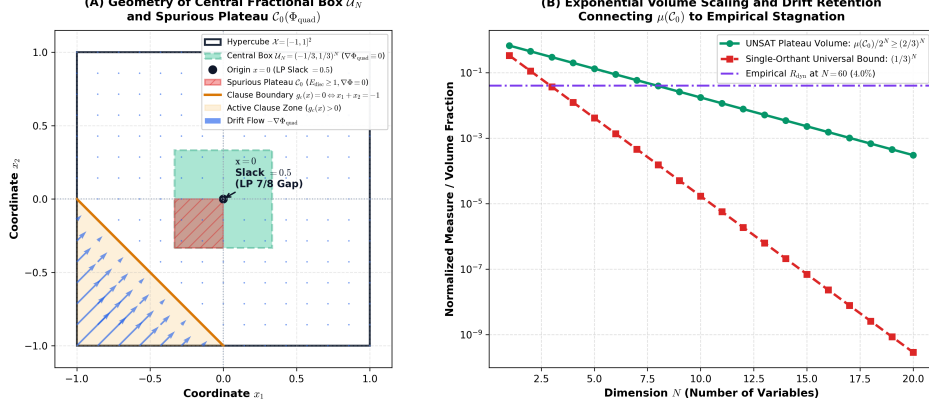


Figure 1: **Illustration of Theorem 1:** The central fractional box $\mathcal{U}_N = (-1/3, 1/3)^N$ embedded inside the hypercube $\mathcal{X} = [-1, 1]^N$. Throughout $\mathcal{U}_N$, every clause violation satisfies $g_c(x) < 0$, creating an exact zero-gradient flat plateau with 0.5 LP interior slack.

3 Theorem 2: Measure Zero of Critical Sets in Analytic Relaxations

Theorem 3.1 (Measure Zero of Critical Sets under (H3')). *Under hypotheses (H1), (H2), and (H3'), the critical zero-sets of the multilinear and Softplus relaxations satisfy:*

$$\mu(\mathcal{C}_0(\Phi_{\text{mult}})) = 0 \quad \text{and} \quad \mu(\mathcal{C}_0(\Phi_{\text{soft}})) = 0. \quad (12)$$

Proof. Case 1: Multilinear (Φ_{mult}). Under (H3'), $\Phi_{\text{mult}} \not\equiv \text{const.}$ Thus there exists an index $k \in \{1, \dots, N\}$ such that $P_k(x) \equiv \partial_k \Phi_{\text{mult}}(x) \not\equiv 0$. By Fubini's theorem and induction on dimension (Okamoto's Lemma [5, 24]), the zero set $Z(P_k) = \{x \in \mathbb{R}^N \mid P_k(x) = 0\}$ has Lebesgue measure zero: $\mu(Z(P_k)) = 0$. Since $\mathcal{C}_0(\Phi_{\text{mult}}) \subseteq Z(\nabla \Phi_{\text{mult}}) \subseteq Z(P_k)$, it follows that $\mu(\mathcal{C}_0(\Phi_{\text{mult}})) = 0$.

Case 2: Softplus (Φ_{soft}). The function $\Phi_{\text{soft}}(x)$ is real analytic ($\mathcal{C}^\omega$) on $\mathbb{R}^N$. By Theorem 6.1, its Laplacian satisfies $\Delta \Phi_{\text{soft}}(x) = \text{Tr}(V^T W(x) V) = \frac{3}{4} \sum_{c=1}^M w_c(x) > 0$ everywhere on $\mathbb{R}^N$ for any formula with $M \geq 1$. Being strictly subharmonic, Φ_{soft} is non-constant. Hence at least one partial derivative $\partial_k \Phi_{\text{soft}}$ is a nonzero real-analytic function. By the zero-set theorem for nontrivial real-analytic functions [14], $\mu(Z(\partial_k \Phi_{\text{soft}})) = 0$. Since $\mathcal{C}_0(\Phi_{\text{soft}}) \subseteq Z(\partial_k \Phi_{\text{soft}})$, the critical set has measure zero. $\square$

4 Theorem 3: Harmonic Multilinear Landscapes & Morse Saddles

Theorem 4.1 (The Harmonic Saddle Theorem). *For any 3-CNF formula F satisfying (H1) and (H3'), the multilinear potential Φ_{mult} has identically vanishing Laplacian:*

$$\Delta \Phi_{\text{mult}}(x) = \text{Tr}(\nabla^2 \Phi_{\text{mult}}(x)) \equiv 0, \quad \forall x \in \mathbb{R}^N. \quad (13)$$

Consequently:

1. Φ_{mult} admits no local minimum in $\text{int}(\mathcal{X})$.
2. Every non-degenerate interior critical point is strictly a hyperbolic saddle of Morse index $1 \leq m \leq N - 1$.

3. For every interior critical point $x^* \in \text{int}(\mathcal{X})$ (whether non-degenerate or degenerate):

$$\forall \varepsilon > 0, \exists y \in \text{int}(\mathcal{X}), \|y - x^*\| < \varepsilon \quad \text{such that} \quad \Phi_{\text{mult}}(y) < \Phi_{\text{mult}}(x^*). \quad (14)$$

Proof. For any clause c , the variables are distinct by (H1). Thus $\frac{\partial^2 \phi_c}{\partial x_i^2} \equiv 0$, yielding $\Delta \Phi_{\text{mult}}(x) = \sum_{i=1}^N 0 \equiv 0$. By the **Strong Minimum Principle for Harmonic Functions** [7, 9], a non-constant harmonic function cannot attain a local minimum in the interior. In non-degenerate critical points, $\text{Tr}(H) = 0 \implies \lambda_1 < 0$ and $\lambda_N > 0$, giving Morse index $1 \leq m \leq N - 1$. For degenerate points, the non-existence of a local minimum implies the existence of lower-energy points in every open ball, reinforced by Milnor's Curve Selection Lemma [20]. $\square$

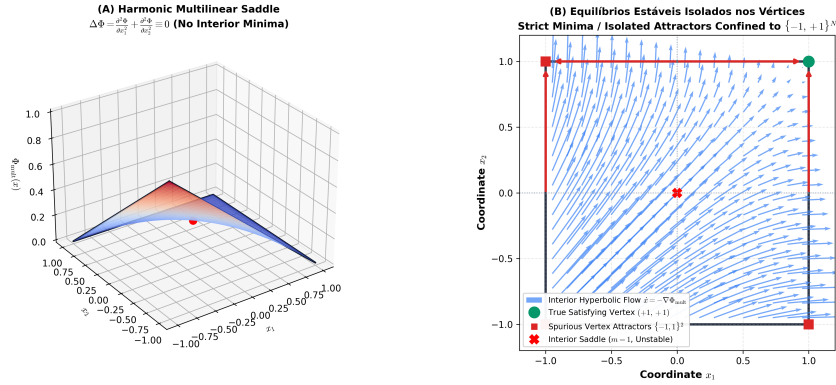


Figure 2: **Illustration of Theorems 3 and 4A'/4B:** The multilinear potential Φ_{mult} is harmonic ($\Delta \Phi_{\text{mult}} \equiv 0$). Every non-degenerate interior critical point is a hyperbolic saddle; degenerate interior points still have lower-energy points in every neighborhood. Under projected gradient flow, all isolated asymptotically stable equilibria are confined to the discrete vertices $\{-1, +1\}^N$, while non-strict attracting continua can occur on the boundary.

5 Theorem 4A' and Corollary 4B: Stratified Hypercube Dynamics and Vertex Confinement (Without Hypothesis H4)

Theorem 5.1 (Theorem 4A': Relative Face Local Minima Inherit Vertex Values). *Under (H1), for any 3-CNF formula, let $x^* \in \text{relint}(\mathcal{F})$ be a local minimum of the restriction of Φ_{mult} relative to a face $\mathcal{F}$ of dimension $d \geq 0$. Then:*

$$\Phi_{\text{mult}}(x^*) = E_{\text{disc}}(v), \quad \forall v \in \mathcal{V}(\mathcal{F}) \quad (15)$$

where $\mathcal{V}(\mathcal{F})$ is the set of Boolean vertices of $\mathcal{F}$. In particular, if x^* is a strict local minimum of $\Phi_{\text{mult}}|_{\mathcal{X}}$ relative to the hypercube, then x^* is strictly a discrete vertex $x^* \in \{-1, +1\}^N$ (face of dimension $d = 0$).

Proof. The hypercube decomposes into disjoint open faces $\mathcal{X} = \bigcup_{\mathcal{F}} \text{relint}(\mathcal{F})$. On any face $\mathcal{F}$ of dimension $d \geq 1$, fixing $N - d$ coordinates to ± 1 preserves multilinearity. The intrinsic face Laplacian vanishes identically: $\Delta_{\mathcal{F}} \Phi_{\mathcal{F}} \equiv 0$. If $x^* \in \text{relint}(\mathcal{F})$ is a local minimum of $\Phi_{\mathcal{F}}$ on the connected open domain $\text{relint}(\mathcal{F})$, then by the Strong Minimum Principle for harmonic functions, $\Phi_{\mathcal{F}}$ must be constant on a neighborhood of x^* . Since $\Phi_{\mathcal{F}}$ is a multilinear polynomial, it is identically constant on the entire compact face $\mathcal{F}$:

$$\Phi_{\text{mult}}(x) \equiv \Phi_{\text{mult}}(x^*), \quad \forall x \in \mathcal{F}. \quad (16)$$

Since all Boolean vertices $\mathcal{V}(\mathcal{F}) \subset \{-1, +1\}^N$ belong to $\mathcal{F}$, and $\Phi_{\text{mult}}(v) = E_{\text{disc}}(v)$, it follows that $\Phi_{\text{mult}}(x^*) = E_{\text{disc}}(v)$ for all $v \in \mathcal{V}(\mathcal{F})$. Consequently, no strict local minimum can exist on any face of dimension $d \geq 1$. All strict local minima reside exclusively in faces of dimension $d = 0$, i.e., discrete vertices $\{-1, +1\}^N$. $\square$

Corollary 5.2 (Corollary 4B: Confinement of Projected Attractors via LaSalle). *Under projected gradient flow $\dot{x}(t) = \Pi_{T_{\mathcal{X}}(x(t))}(-\nabla\Phi_{\text{mult}}(x(t)))$, every isolated asymptotically stable equilibrium is strictly a discrete vertex $x^* \in \{-1, +1\}^N$.*

Proof. Consider the Lyapunov function $V(x) = \Phi_{\text{mult}}(x)$. By Moreau's projection property on closed convex cones:

$$\dot{V}(x(t)) = \langle \nabla\Phi_{\text{mult}}(x), \Pi_{T_{\mathcal{X}}(x)}(-\nabla\Phi_{\text{mult}}(x)) \rangle = -\|\Pi_{T_{\mathcal{X}}(x)}(-\nabla\Phi_{\text{mult}}(x))\|^2 \leq 0. \quad (17)$$

By LaSalle's Invariance Principle for projected dynamical systems [3, 22], every omega-limit set on the compact set $\mathcal{X}$ is contained in the largest invariant subset of $\{\Pi_{T_{\mathcal{X}}(x)}(-\nabla\Phi) = 0\}$; equivalently, trajectories approach the projected stationary set $\{-\nabla\Phi(x^*) \in N_{\mathcal{X}}(x^*)\}$. An isolated asymptotically stable equilibrium x^* must be a strict local minimum of Φ_{mult} relative to $\mathcal{X}$: if arbitrarily close points had strictly smaller energy, monotonicity of Φ_{mult} would prevent their trajectories from converging upward to x^* ; if arbitrarily close distinct points had equal energy, then either their trajectories immediately decrease below $\Phi(x^*)$ or they are additional equilibria, contradicting isolation. By Theorem 5.1, no point in the relative interior of a face $\mathcal{F}$ with $\dim(\mathcal{F}) \geq 1$ can be a strict local minimum. Hence all isolated asymptotically stable equilibria belong to faces of dimension $d = 0$: $x^* \in \{-1, +1\}^N$. $\square$

6 Theorem 5: Hessian Factorization and Spectral Conditioning

Define the **Clause Incidence Matrix** $V \in \mathbb{R}^{M \times N}$ with rows $v_c^T = -\frac{1}{2}(\sigma^{(c)})^T$:

$$V = -\frac{1}{2} \begin{pmatrix} \sigma_1^{(1)} & \dots & \sigma_N^{(1)} \\ \vdots & \ddots & \vdots \\ \sigma_1^{(M)} & \dots & \sigma_N^{(M)} \end{pmatrix} \in \mathbb{R}^{M \times N}. \quad (18)$$

Theorem 6.1 (Hessian Factorization and Spectral Bounds). *For the Softplus potential $\Phi_{\text{soft}}(x) = \sum_{c=1}^M \frac{1}{\beta} \ln(1 + e^{\beta g_c(x)})$:*

1. *The Hessian admits the exact global factorization:*

$$\nabla^2 \Phi_{\text{soft}}(x) = V^T W(x) V \succeq 0, \quad (19)$$

where $W(x) = \text{diag}(w_1(x), \dots, w_M(x))$ with strictly positive weights:

$$w_c(x) = \beta \sigma(\beta g_c(x)) (1 - \sigma(\beta g_c(x))) > 0. \quad (20)$$

2. *Φ_{soft} is globally convex on $\mathbb{R}^N$. If $\text{rank}(V) = N$, then $\nabla^2 \Phi_{\text{soft}}(x) \succ 0$ and Φ_{soft} is strictly convex.*
3. *If $\text{rank}(V) = N$ (so that $V^T V \succ 0$), the spectral condition number $\kappa(\nabla^2 \Phi_{\text{soft}}(x))$ is bounded by:*

$$\kappa(\nabla^2 \Phi_{\text{soft}}(x)) \leq \frac{\max_c w_c(x)}{\min_c w_c(x)} \cdot \frac{\lambda_{\max}(V^T V)}{\lambda_{\min}(V^T V)} = \kappa(W(x)) \cdot \kappa(V^T V). \quad (21)$$

Proof. Let $v_c = -\frac{1}{2}\sigma^{(c)}$. Since $g_c(x) = v_c^T x - 1/2$, we have $\nabla g_c = v_c$ and $\nabla^2 g_c \equiv 0$. Moreover,

$$\nabla \Phi_{\text{soft}}(x) = \sum_{c=1}^M \sigma(\beta g_c(x)) v_c = V^T \sigma(\beta g(x)),$$

and differentiation gives

$$\nabla^2 \Phi_{\text{soft}}(x) = \sum_{c=1}^M \beta \sigma(\beta g_c) (1 - \sigma(\beta g_c)) v_c v_c^T = V^T W(x) V.$$

For any $z \in \mathbb{R}^N$,

$$z^T \nabla^2 \Phi_{\text{soft}}(x) z = \|W(x)^{1/2} V z\|_2^2 \geq 0,$$

which establishes convexity. Courant–Fischer gives

$$\lambda_{\min}(H) \geq (\min_c w_c) \lambda_{\min}(V^T V), \quad \lambda_{\max}(H) \leq (\max_c w_c) \lambda_{\max}(V^T V),$$

yielding the condition-number bound. $\square$

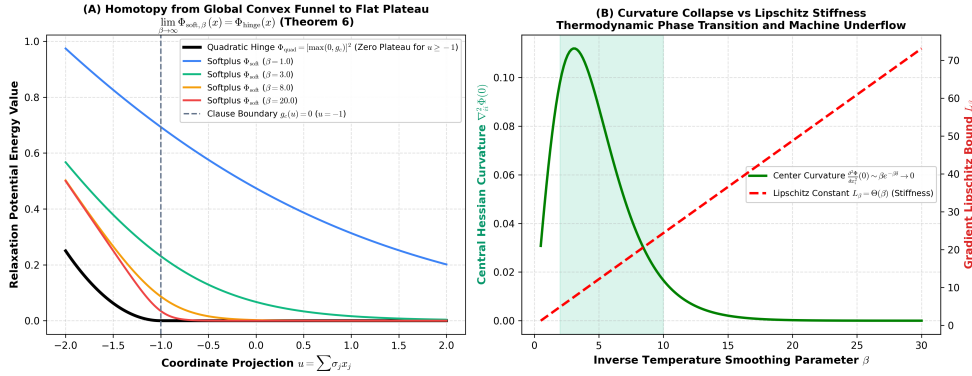


Figure 3: **Illustration of Theorems 5 and 6:** The Softplus relaxation Hessian factors as $\nabla^2 \Phi_{\text{soft}} = V^T W(x) V \succeq 0$, creating a strictly convex funnel when $\text{rank}(V) = N$. As $\beta \rightarrow \infty$, the gradient-field Lipschitz constant scales as $L_\beta = \Theta(\beta)$, while finite-precision evaluation can drive the exponentially small Softplus gradients into subnormal/underflow regimes, numerically approaching the flat Hinge plateau.

7 Theorem 6: Lipschitz Gradient Bounds and IEEE 754 Underflow

Theorem 7.1 (Lipschitz Constant of Gradient Field and Numerical Underflow). *For the Softplus relaxation as $\beta \rightarrow \infty$, let $d_{\max} = \max_i |\{c : i \in c\}|$ be the maximum variable degree:*

1. The **Lipschitz constant of the gradient field** $L_\beta \equiv \sup_x \|\nabla^2 \Phi_{\text{soft}}(x)\|_2 = \Theta(\beta)$ satisfies:

$$\frac{3}{16}\beta \leq L_\beta \leq \frac{3d_{\max}}{16}\beta. \quad (22)$$

2. On the contracted sub-box $\mathcal{U}_N(\rho) = (-\rho, \rho)^N$ with $\rho < 1/3$:

$$\|\nabla \Phi_{\text{soft}}(x)\|_\infty \leq \frac{M}{2} e^{-\beta(1-3\rho)/2} \quad \text{and} \quad \|\nabla \Phi_{\text{soft}}(x)\|_2 \leq \frac{\sqrt{3}M}{2} e^{-\beta(1-3\rho)/2}. \quad (23)$$

3. At the origin $x = \mathbf{0}$, the idealized exponential scale $e^{-\beta/2}$ crosses the IEEE 754 binary32 minimum normal, minimum subnormal, and round-to-nearest zero thresholds at approximately $\beta = 174.67, 206.56$, and 207.94 , respectively; for binary64 the corresponding values are $\beta = 1416.79, 1488.88$, and 1490.27 . These are scales for the exponential factor itself, not implementation-independent thresholds for a particular sigmoid routine; FTZ/DAZ modes can zero subnormals earlier.

Proof. Since $0 < w_c(x) \leq \beta/4$,

$$\|\nabla^2 \Phi_{\text{soft}}(x)\|_2 \leq \frac{\beta}{4} \lambda_{\max}(V^T V).$$

Each row of V has exactly three nonzero entries of magnitude $1/2$. For a fixed variable i , the absolute row sum of $V^T V$ is at most $3d_i/4$, hence Gershgorin gives $\lambda_{\max}(V^T V) \leq 3d_{\max}/4$. Therefore $L_\beta \leq 3d_{\max}\beta/16$.

For the lower bound, choose a clause c and a point with $g_c(x) = 0$. Then $w_c = \beta/4$. Since $\|v_c\|^2 = 3/4$ and all Hessian summands are positive semidefinite, $\lambda_{\max}(\nabla^2 \Phi_{\text{soft}}) \geq (\beta/4)(3/4) = 3\beta/16$.

If $x \in \mathcal{U}_N(\rho)$, then every clause obeys $g_c(x) \leq -(1 - 3\rho)/2$. Hence $\sigma(\beta g_c) \leq e^{-\beta(1-3\rho)/2}$. Using $\|v_c\|_\infty = 1/2$ and $\|v_c\|_2 = \sqrt{3}/2$ and the triangle inequality yields the stated gradient bounds.

At $x = 0$, $g_c(0) = -1/2$, so the relevant idealized exponential scale is $e^{-\beta/2}$. Equating it with 2^{-126} , 2^{-149} , and 2^{-150} gives the binary32 values 174.67, 206.56, and 207.94; equating it with 2^{-1022} , 2^{-1074} , and 2^{-1075} gives the binary64 values 1416.79, 1488.88, and 1490.27. The last value in each triple is the ideal round-to-nearest transition below half the smallest subnormal. Actual evaluation of the logistic function is implementation dependent, and FTZ/DAZ modes may zero subnormals earlier. $\square$

8 Theorem 7B and Proposition 7A: Universal Centripetal Contraction

Theorem 8.1 (Theorem 7B: Universal Centripetal Contraction of Quadratic Hinge). *For any 3-CNF formula and any point $x \in \mathcal{X}$ with active clauses $\text{act}(x) = \{c \mid g_c(x) > 0\} \neq \emptyset$:*

1. *The gradient field satisfies strict centripetal contraction:*

$$\langle -\nabla \Phi_{\text{quad}}(x), x \rangle = - \sum_{c \in \text{act}(x)} (2g_c(x)^2 + g_c(x)) < 0. \quad (24)$$

2. *The squared norm $\|x(t)\|_2^2$ is a strict Lyapunov function outside Z :*

$$\frac{d}{dt} \|x(t)\|_2^2 = 2\langle x(t), \dot{x}(t) \rangle \leq 2\langle x(t), -\nabla \Phi_{\text{quad}}(x(t)) \rangle < 0. \quad (25)$$

3. *The set of projected equilibria $\mathcal{E}_{\text{proj}} = \{x \in \mathcal{X} \mid -\nabla \Phi_{\text{quad}}(x) \in N_{\mathcal{X}}(x)\}$ coincides identically with the LP relaxation polytope:*

$$\mathcal{E}_{\text{proj}} \equiv Z. \quad (26)$$

4. *By LaSalle's Invariance Principle on the compact hypercube $\mathcal{X} = [-1, 1]^N$, every trajectory under projected gradient flow approaches the LP equilibrium polytope $Z = \{x \in \mathcal{X} \mid g_c(x) \leq 0, \forall c\}$, i.e., $\lim_{t \rightarrow \infty} \text{dist}(x(t), Z) = 0$.*

Proof. Let $\mathcal{X} = [-1, 1]^N$. At any point $x \in \mathcal{X}$, the outward normal cone in the sense of convex analysis is:

$$N_{\mathcal{X}}(x) = \left\{ \nu \in \mathbb{R}^N \mid \nu_i = 0 \text{ if } |x_i| < 1, \quad \nu_i x_i \geq 0 \text{ if } |x_i| = 1 \right\}. \quad (27)$$

The tangent cone is $T_{\mathcal{X}}(x) = \{w \in \mathbb{R}^N \mid \langle w, \nu \rangle \leq 0, \forall \nu \in N_{\mathcal{X}}(x)\}$, and the projected gradient dynamical system is defined by $\dot{x} = \Pi_{T_{\mathcal{X}}(x)}(-\nabla \Phi_{\text{quad}}(x))$, where $\Pi_{T_{\mathcal{X}}(x)}$ is the orthogonal projection operator onto the tangent cone:

$$[\Pi_{T_{\mathcal{X}}(x)}(v)]_i = \begin{cases} v_i, & \text{if } |x_i| < 1 \text{ or } (x_i = +1, v_i \leq 0) \text{ or } (x_i = -1, v_i \geq 0), \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

A state $x^* \in \mathcal{X}$ is a projected equilibrium if $\dot{x} = \mathbf{0}$, which is geometrically equivalent to $-\nabla \Phi_{\text{quad}}(x^*) \in N_{\mathcal{X}}(x^*)$.

For active c , $g_c(x) > 0 \iff \sigma^{(c)} \cdot x = -(2g_c(x) + 1)$. The unconstrained gradient is $-\nabla \Phi_{\text{quad}}(x) = \sum_{c \in \text{act}} g_c(x) \sigma^{(c)}$. Taking the Euclidean inner product with x :

$$\langle -\nabla \Phi_{\text{quad}}(x), x \rangle = \sum_{c \in \text{act}} g_c(x) (\sigma^{(c)} \cdot x) = - \sum_{c \in \text{act}} (2g_c(x)^2 + g_c(x)) < 0. \quad (29)$$

Writing $v = -\nabla \Phi_{\text{quad}}(x) = d + n$ by Moreau decomposition, with $d = \Pi_{T_{\mathcal{X}}(x)}v$ and $n \in N_{\mathcal{X}}(x)$, the box normal-cone sign condition gives $\langle x, n \rangle \geq 0$. Therefore $\langle x, d \rangle \leq \langle x, v \rangle < 0$ whenever an active clause exists, which proves the strict decrease of $\|x(t)\|^2$ asserted in item 2. We now verify the exact identity of projected equilibria $\mathcal{E}_{\text{proj}} \equiv Z$:

1. If $x \in Z$, then $g_c(x) \leq 0$ for all c , so $\text{act}(x) = \emptyset$ and $-\nabla \Phi_{\text{quad}}(x) = \mathbf{0} \in N_{\mathcal{X}}(x)$ for all $x \in \mathcal{X}$, yielding $Z \subseteq \mathcal{E}_{\text{proj}}$.
2. Conversely, suppose $x \notin Z$. Then $\text{act}(x) \neq \emptyset$, which guarantees $\langle -\nabla \Phi_{\text{quad}}(x), x \rangle < 0$. On the other hand, for any vector $\nu \in N_{\mathcal{X}}(x)$, the sign condition $\nu_i x_i = |\nu_i| \geq 0$ ensures:

$$\langle \nu, x \rangle = \sum_{i: |x_i|=1} \nu_i x_i = \sum_{i: |x_i|=1} |\nu_i| \geq 0. \quad (30)$$

Consequently, $-\nabla \Phi_{\text{quad}}(x)$ cannot belong to $N_{\mathcal{X}}(x)$, since any element $\nu \in N_{\mathcal{X}}(x)$ must satisfy $\langle \nu, x \rangle \geq 0$, whereas $\langle -\nabla \Phi_{\text{quad}}(x), x \rangle < 0$. This establishes that no projected equilibria exist outside Z , neither in the interior nor on the boundary faces: $\mathcal{E}_{\text{proj}} \subseteq Z$.

Therefore, $\mathcal{E}_{\text{proj}} \equiv Z$ holds identically on $\mathcal{X}$. Since the energy dissipation satisfies $\dot{\Phi}_{\text{quad}}(x) = -\|\Pi_{T_{\mathcal{X}}(x)}(-\nabla \Phi_{\text{quad}}(x))\|_2^2 \leq 0$, by LaSalle's Invariance Principle all trajectories approach the largest invariant set contained in $\{\dot{\Phi}_{\text{quad}} = 0\} = \mathcal{E}_{\text{proj}}$, which is identically Z , establishing $\lim_{t \rightarrow \infty} \text{dist}(x(t), Z) = 0$. $\square$

Remark 8.2 (Clarification on Unsatisfiable Formulas). *For any unsatisfiable (UNSAT) formula, every Boolean vertex satisfies $E_{\text{disc}}(s) \geq 1$. Consequently, the spurious basin mass is identically $\mathcal{M}_{\text{spur}} \equiv 1$ for all continuous representations $(\Phi_{\text{quad}}, \Phi_{\text{mult}}, \Phi_{\text{soft}})$. The separation $\mathcal{M}_{\text{spur}}(\Phi_{\text{quad}}) - \mathcal{M}_{\text{spur}}(\Phi_{\text{mult}}) = 1 - 1 = 0$ is zero for UNSAT. The true divergence occurs on satisfiable instances where Φ_{mult} finds satisfying vertices while Φ_{quad} is trapped in the fractional LP polytope Z .*

Proposition 8.3 (Proposition 7A: Competitive Jacobian Structure for Purely Negative Clause Families (Under Re-Audit)). *Let $N \geq 4$. Consider the formula F_N consisting of all $M = \binom{N}{3}$ purely negative clauses $c = (\neg x_i \vee \neg x_j \vee \neg x_k)$.*

1. **Competitive Jacobian Structure:** For the multilinear relaxation Φ_{mult} , each clause potential is $P_c(x) = \left(\frac{1+x_i}{2}\right)\left(\frac{1+x_j}{2}\right)\left(\frac{1+x_k}{2}\right)$. On the hypercube $\mathcal{X} = [-1, 1]^N$, the cross-derivatives satisfy $\frac{\partial^2 P_c}{\partial x_i \partial x_j} = \frac{1+x_k}{8} \geq 0$. Consequently, the off-diagonal entries of the Jacobian matrix $J(x) = D(-\nabla \Phi_{\text{mult}}(x)) = -\nabla^2 \Phi_{\text{mult}}(x)$ satisfy:

$$J_{ij}(x) = -\frac{\partial^2 \Phi_{\text{mult}}}{\partial x_i \partial x_j}(x) = -\sum_{c \ni \{i,j\}} \frac{\partial^2 P_c}{\partial x_i \partial x_j}(x) \leq 0, \quad \forall i \neq j. \quad (31)$$

2. **Dynamical Consequence and Suspension of Hirsch:** The continuous multilinear gradient flow $\dot{x} = -\nabla \Phi_{\text{mult}}(x)$ is competitive (weakly inhibitory) with non-positive off-diagonal entries ($J_{ij}(x) \leq 0$) on $\mathcal{X} = [-1, 1]^N$. It violates the Kamke-Müller condition ($J_{ij} \geq 0$) required for cooperative/monotone dynamical systems, precluding any direct invocation of Hirsch's monotone convergence theorems.
3. **Formal Status:** The dynamical characterization of the multilinear flow on F_N is formally suspended under active structural re-audit.

Remark 8.4 (Analytical Audit Note on Proposition 7A). The derivation confirms that interactions between variables in purely negative clauses exert mutual competitive (weakly inhibitory) suppression ($J_{ij} \leq 0$). Analyzing convergence on such competitive systems in dimension $N \geq 3$ requires carrying simplices and invariant manifold techniques rather than order-preserving flows.

9 Theorem 8: Finite-Dimensional Lower Bound on LP Polytope Volume via Jensen's Inequality

Theorem 9.1 (Theorem 8: Finite-Dimensional Lower Bound on LP Polytope Volume via Jensen's Inequality). For the independent-clause ensemble $\mathcal{E}_{\text{iid}}(N, \alpha)$ with $M = \lfloor \alpha N \rfloor$: For every finite dimension $N \geq 3$, the expected normalized volume of the LP polytope $Z = \{x \in [-1, 1]^N \mid g_c(x) \leq 0, \forall c\}$ satisfies the rigorous finite-dimensional exponential lower bound:

$$\mathbb{E}[\mu_{\text{norm}}(Z)] \geq \left(\frac{5}{6}\right)^{\lfloor \alpha N \rfloor} \geq \exp\left(-N\alpha \ln\left(\frac{6}{5}\right)\right) > 0. \quad (32)$$

Moreover, $\mathbb{E}[\mu_{\text{norm}}(Z)] \geq \mu_{\text{norm}}(\mathcal{U}_N) = (1/3)^N > 0$.

Proof. By Fubini's theorem:

$$\mathbb{E}_F[\mu_{\text{norm}}(Z)] = \mathbb{E}_F \left[\int_{[-1, 1]^N} \prod_{c=1}^M \mathbf{1}_{\{g_c(x) \leq 0\}} \frac{dx}{2^N} \right] = \int_{[-1, 1]^N} [p(x)]^M \frac{dx}{2^N}, \quad (33)$$

where $p(x) = \mathbb{P}_c(g_c(x) \leq 0)$. For a single clause under $x \sim \text{Unif}([-1, 1]^N)$, $u_1 + u_2 + u_3 \geq 1$ for $u_j \sim \text{Unif}([0, 1])$. By the Irwin-Hall distribution of order 3, $\mathbb{P}(S_3 < 1) = 1/6 \implies \int p(x) \frac{dx}{2^N} = 5/6$. For $M = \lfloor \alpha N \rfloor \geq 1$, by Jensen's inequality for the convex function $t \mapsto t^M$ on $[0, 1]$:

$$\mathbb{E}_F[\mu_{\text{norm}}(Z)] \geq \left(\int_{[-1, 1]^N} p(x) \frac{dx}{2^N} \right)^M = \left(\frac{5}{6}\right)^M = \left(\frac{5}{6}\right)^{\lfloor \alpha N \rfloor} \geq \left(\frac{5}{6}\right)^{\alpha N} > 0. \quad (34)$$

This establishes a strictly positive lower bound for each finite dimension N . We emphasize that this lower bound does not demonstrate that $\lim_{N \rightarrow \infty} \mathbb{E}[\mu_{\text{norm}}(Z)] = 0$, which would require an independent upper bound. $\square$

10 Theorems 9 and 10: Dynamic Analysis, Audit, and Refutation

Lemma 10.1 (Lemma 9.1: Implication Chain Dynamics, Stirling Expansion, and Unit-Fact Falsification). *Consider an implication chain of length $K \geq 2$, $x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_K$ ($\neg x_k \vee x_{k+1}$):*

1. **Pure Implication Chain (Conservation of Mean vs. Isotonic Limit):** *In the absence of external facts, the Hinge penalty is*

$$\Phi_{\text{quad}}(x) = \sum_{k=1}^{K-1} \left[\max\left(0, \frac{x_k - x_{k+1}}{2}\right) \right]^2.$$

Trajectories strictly conserve the center of mass,

$$\frac{d}{dt} \sum_{k=1}^K x_k(t) \equiv 0,$$

constraining the dynamics to affine hyperplanes of constant mean. While heuristic and numerical evidence associates the asymptotic state with the Euclidean Isotonic Projection $\Pi_Z(x_0)$ (via Pool Adjacent Violators) onto the cone $Z = \{x \in [-1, 1]^K \mid x_1 \leq x_2 \leq \dots \leq x_K\}$, the rigorous proof that continuous non-smooth Hinge gradient flow trajectories converge identically to $\Pi_Z(x_0)$ constitutes an independent analytical problem and remains open.

2. **Conditional Sparre–Andersen Calculation:** *If the still-unproved identification $x^* = \Pi_Z(x_0)$ with the Euclidean isotonic projection is assumed, then $x_1^* > 0$ is equivalent to positivity of all prefix sums of the i.i.d. symmetric continuous increments. Sparre–Andersen then gives*

$$\mathbb{P}(x_1^* > 0 \mid x^* = \Pi_Z(x_0)) = \frac{\binom{2K}{K}}{4^K} = \frac{1}{\sqrt{\pi K}} \left(1 - \frac{1}{8K} + \mathcal{O}(K^{-2}) \right). \quad (35)$$

This calculation is conditional evidence, not an unconditional theorem about the continuous Hinge flow.

3. **Falsification of Spurious Trapping under Positive Unit Fact:** *If a positive unit fact $x_1 = 1$ is incorporated via penalty $h(x_1) = [\max(0, (1 - x_1)/2)]^2$, its gradient injects a strictly positive upward force $-\frac{\partial h}{\partial x_1} = \frac{1-x_1}{2} > 0$ for $x_1 < 1$. This breaks the conservation of the center of mass ($\frac{d}{dt} \sum x_k > 0$). Furthermore, the zero set of the resulting convex squared-Hinge objective collapses to the singleton $Z = \{(+1, \dots, +1)\}$. Standard projected convex-gradient-flow convergence therefore forces every trajectory to approach that singleton. Thus the former claim of a spurious trapping basin of mass $1 - o(1)$ for chains with a positive head fact is false; numerical integration is only a consistency check, not the proof.*

Theorem 10.2 (Theorem 9: Dynamical Behavior in Monotone Horn-3-SAT — Open Status). *Consider monotone Horn-3-SAT formulas F_N where all-negative assignments $s = (-1, \dots, -1)$ are satisfying:*

1. *Isolated binary implications $\neg x_i \vee x_j$ and purely negative clauses exhibit different Jacobian sign patterns. More generally, a Horn clause such as $(\neg x_1 \vee \neg x_2 \vee x_3)$ has $J_{12} = -(1 - x_3)/8 \leq 0$ but $J_{13} = (1 + x_2)/8 \geq 0$ and $J_{23} = (1 + x_1)/8 \geq 0$. Thus general Horn 3-SAT has sign-indefinite off-diagonal interactions and is not, in general, a cooperative or competitive system with respect to the standard positive orthant order. Proposition 8.3 applies only to the purely negative family. Standard Hirsch monotonicity results therefore do not directly settle the general Horn case, and global convergence to satisfying models remains open.*

2. **Status of Dynamic Separation:** The exact spurious basin mass of Φ_{quad} for general Horn DAGs remains an **Open Problem**. In particular, single linear chains with positive head facts cannot establish the desired separation because their zero set collapses to the singleton

$$Z = \{(+1, \dots, +1)\}.$$

Lemma 10.3 (Lemma 10.1: Subcritical Hypergraph Component Decoupling, Degree Identity, and Defect Tightness). *For random 3-SAT below the subcritical threshold $\alpha < \alpha_c = 1/6$:*

1. **Subcritical Branching and Connected Component Decoupling:** The clause exploration branching factor satisfies $R_0 = 2 \times 3\alpha = 6\alpha < 1$. Consequently, the formula hypergraph F_N partitions into a collection of mutually disjoint connected components $F_N = \bigcup_j \mathcal{K}_j$, where distinct components share no variables ($V(\mathcal{K}_j) \cap V(\mathcal{K}_{j'}) = \emptyset$ for $j \neq j'$). The continuous multilinear energy is strictly separable across components:

$$\Phi_{\text{mult}}(x) = \sum_j \Phi_{\mathcal{K}_j}(x_{V(\mathcal{K}_j)}), \quad (36)$$

and the projected gradient dynamical system decouples into mutually independent autonomous flows $\dot{x}_{V(\mathcal{K}_j)} = \Pi_{[-1,1]^{V(\mathcal{K}_j)}}(-\nabla \Phi_{\mathcal{K}_j}(x_{V(\mathcal{K}_j)}))$.

2. **Global Degree Identity on Tree Components:** Let $\mathcal{C}_{\text{tree}}$ denote the class of connected components $\mathcal{K}_j$ that are acyclic (linear hypertrees). Because variables in $V(\mathcal{K}_j)$ appear in zero clauses outside $\mathcal{K}_j$, the degree of any variable in $\mathcal{K}_j$ is **identically equal to its global degree** in the original formula:

$$\deg_{F_N}(v) \equiv \deg_{\mathcal{K}_j}(v), \quad \forall v \in V(\mathcal{K}_j). \quad (37)$$

Recursive leaf peeling eliminates every hyperedge of a linear hypertree, but this combinatorial fact does not imply that every positive clause at a projected equilibrium contains a degree-one variable. A four-clause star already has a positive projected equilibrium on its leafless central clause, and the six-clause construction in Theorem 10.7 has a positive-dimensional attracting equilibrium family. Thus the former universal hypothesis H_{leaf} is withdrawn; only the local-global degree identity itself remains valid.

3. **Defect Tightness and Negligible Density (Strategy B):** Let $\mathcal{C}_{\text{defect}}$ denote the collection of connected components containing cycles (either clause pairs sharing ≥ 2 variables or loose cycles of length $k \geq 3$). Since $R_0 = 6\alpha < 1$, the expected number of cycles of all orders is $\mathcal{O}(1)$ and component sizes exhibit an exponential tail. Thus, the total number of clauses in defective components satisfies $\mathbb{E}[M_{\text{defect}}] = \mathcal{O}(1)$, ensuring $M_{\text{defect}} = \mathcal{O}_{\mathbb{P}}(1)$ (tightness). Even under the worst-case adversarial assumption that every clause in defective components is violated, the multilinear discrete energy contribution from defects is bounded by $M_{\text{defect}} = \mathcal{O}_{\mathbb{P}}(1)$.

Lemma 10.4 (Former Lemma 10.2: Universal Strict-Saddle Claim — Refuted). *Coordinate-wise multilinearity still gives $\text{Tr}(\mathcal{H}_{\mathcal{F}}) = 0$ on every positive-dimensional face. The former proof, however, inferred $\mathcal{H}_{\mathcal{F}} \neq 0$ from an unproved “violated leaf clause.” That inference is false. The six-clause linear hypertree in Theorem 10.7 has a nine-dimensional family of positive non-strict relative minima with an open basin. Hence neither the universal strict-saddle conclusion nor the associated almost-sure evasion claim is valid.*

Lemma 10.5 (Former Lemma 10.3: Universal Transverse Repulsion — Insufficient). *The transverse calculation for an edge that actually has the stipulated exclusive leaf remains a valid local statement. It is not universal on tree components, because the existence of that leaf in a positive clause does not follow from acyclicity. More importantly, excluding equilibria from one-dimensional open edges cannot exclude attracting equilibrium families on higher-dimensional faces. This lemma therefore supplies no route to the former conclusion $\rho_{\text{mult}} \rightarrow 0$.*

Lemma 10.6 (Lemma 10.4: Subcritical Hinge Basin Lower Bound and Residual Energy Density). *For the uniform distinct-clause ensemble $\mathcal{E}_{\text{unif}}(N, \alpha)$ with $\alpha < 1/6$, under pure projected gradient flow $\dot{x} = \Pi_{T_{\mathcal{X}}(x)}(-\nabla \Phi_{\text{quad}}(x))$ initialized from $x_0 \sim \text{Unif}([-1, 1]^N)$:*

1. **Decoupled Dynamics on Isolated Clauses:** *Let $\mathcal{C}_{\text{iso}}$ denote the set of isolated clauses in F_N sharing no variables with any other clause. For each $c \in \mathcal{C}_{\text{iso}}$, the projected flow on its variables $(x_{i_1}, x_{i_2}, x_{i_3})$ is strictly autonomous and decoupled from all other coordinates, with initial condition $(x_{0,i_1}, x_{0,i_2}, x_{0,i_3}) \sim \text{Unif}([-1, 1]^3)$ statistically independent across clauses.*
2. **Boundary Non-Saturation (No-Clipping):** *For any initial point outside the LP polytope Z ($S_0 = \sum_{p=1}^3 \sigma_p x_{0,p} < -1$), the trajectory translates linearly at identical rates $\dot{U}_p = g_c(U) > 0$, landing on ∂Z at $U_p^* = U_p(0) + \frac{-1-S_0}{3}$. Because $U_p^* \in (-1, 1)$ strictly for almost all initial conditions, no boundary clipping occurs before reaching ∂Z .*
3. **Exact Rounding Failure Probability:** *The probability that the discrete Boolean assignment $\hat{s} = \text{sign}(x^*)$ violates clause c is given analytically by the sum of two disjoint regions in $[-1, 1]^3$:*

$$\begin{aligned} p_{\text{fail}} &= \mathbb{P}(x_0 \in Z \text{ and } \hat{s} \text{ violates } c) \\ &\quad + \mathbb{P}(x_0 \notin Z \text{ and landing } x^* \text{ violates } c) \\ &= \frac{1}{48} + \frac{7}{96} = \frac{3}{32} = 0.09375. \end{aligned} \tag{38}$$

4. **Density Scaling and Exact Expectation Lower Bound:** *In $\mathcal{E}_{\text{unif}}(N, \alpha)$, the expected number of isolated clauses is $\mathbb{E}[|\mathcal{C}_{\text{iso}}|] = \alpha N e^{-9\alpha}(1 + \mathcal{O}(1/N))$. Since the discrete residual density is normalized by total clauses $M = \alpha N$, the factor α cancels exactly:*

$$\frac{\mathbb{E}[|\mathcal{C}_{\text{iso}}|]}{M} = \frac{\alpha N e^{-9\alpha}(1 + \mathcal{O}(1/N))}{\alpha N} = e^{-9\alpha}(1 + \mathcal{O}(1/N)). \tag{39}$$

Conditionally on the formula, the rounding failure of each isolated clause is an independent Bernoulli trial with parameter $p_{\text{fail}} = 3/32$. Consequently, the expected asymptotic residual density satisfies:

$$\liminf_{N \rightarrow \infty} \mathbb{E}[\rho_{\text{quad}}(\alpha)] \geq \frac{3}{32} e^{-9\alpha} > 0. \tag{40}$$

5. **Asymptotically Almost Sure Concentration:** *By the law of total variance, $\text{Var}(X_{\text{iso}}) = \mathcal{O}(N)$, so the density variance satisfies $\text{Var}(X_{\text{iso}}/M) = \mathcal{O}(1/(\alpha^2 N)) = \mathcal{O}(1/N)$. By Chebyshev's inequality, for any $\varepsilon > 0$:*

$$\lim_{N \rightarrow \infty} \mathbb{P}\left(\rho_{\text{quad}}(\alpha) \geq \frac{3}{32} e^{-9\alpha} - \varepsilon\right) = 1. \tag{41}$$

Theorem 10.7 (Audit Refutation of the Former Theorem 10). *Let $M = \lfloor \alpha N \rfloor$ with fixed $\alpha > 0$, let F be sampled uniformly from the M -subsets of the $Q_N = 8 \binom{N}{3}$ distinct signed 3-SAT clauses, initialize the projected multilinear gradient flow with $x_0 \sim \text{Unif}([-1, 1]^N)$ independently of F , and round its limit coordinatewise to a Boolean assignment. With the notation above,*

$$\liminf_{N \rightarrow \infty} \bar{\rho}_{N, \text{mult}}(\alpha) \geq \frac{729}{2^{59}} \alpha^5 e^{-39\alpha} > 0. \tag{42}$$

Moreover,

$$\mathbb{P}_{F, x_0} \left[\rho_{N, \text{mult}}(F, x_0) \geq \frac{729}{2^{60}} \alpha^5 e^{-39\alpha} \right] \rightarrow 1. \tag{43}$$

In fact, writing

$$d(\alpha) = \frac{729}{64} \alpha^6 e^{-39\alpha}, \quad p_0 = 2^{-53},$$

for all sufficiently large N ,

$$\mathbb{P}_F \left[\mathbb{P}_{x_0} \left(\rho_{N, \text{mult}}(F, x_0) \geq \frac{729}{260} \alpha^5 e^{-39\alpha} \mid F \right) \geq 1 - \exp \left(-\frac{3p_0 d(\alpha)}{128} N \right) \right] \geq 1 - \exp \left(-\frac{d(\alpha)^2}{512\alpha} N \right). \quad (44)$$

The independent-clause model has the same leading expectation constant. Hence the former claim that the multilinear residual tends to zero, both in expectation and with asymptotically high probability, is false for every fixed $\alpha > 0$ and in particular throughout $0 < \alpha < 1/6$.

Proof. Use the signed linear hypertree on vertices $0, \dots, 12$ with clauses

$$\begin{aligned} &(0, 1, 2), (0, 3, 4), (1, 5, 6), \\ &(2, 7, 8), (3, 9, 10), (4, 11, 12), \end{aligned} \quad (45)$$

and sign triples $(+++), (-++), (-+-), (-+-), (-+-), (-+-)$. In affine factor coordinates in $[0, 1]$, its potential is

$$\begin{aligned} \Phi &= yu_1u_2 + (1-y)v_1v_2 + (1-u_1)a_1b_1 + (1-u_2)a_2b_2 \\ &\quad + (1-v_1)c_1d_1 + (1-v_2)c_2d_2. \end{aligned} \quad (46)$$

The quantitative bounds below use the original projected-flow time in $x \in [-1, 1]^N$. If a factor coordinate is $z = (1 \pm x)/2$, then $\partial_x \Phi = \pm \frac{1}{2} \partial_z \Phi$ and $\dot{z} = \pm \frac{1}{2} \dot{x}$; hence, away from the harmless tangent-cone projection notation,

$$\dot{z} = \frac{1}{4} \Pi_{T_{[0,1]}(z)}(-\nabla_z \Phi).$$

Equivalently, with the rescaled time $\tau = t/4$ one recovers $dz/d\tau = \Pi_{T_{[0,1]}(z)}(-\nabla_z \Phi)$. Thus all orbit and basin statements are unchanged by the affine coordinate change, while numerical speed bounds in the original time carry the factor $1/4$. The set

$$\mathcal{M} = \{u_1 = u_2 = v_1 = v_2 = 1, 0 < y < 1, a_j b_j > y, c_j d_j > 1 - y\} \quad (47)$$

is a nine-dimensional family of projected equilibria with $\Phi = 1$. It consists locally of non-strict relative minima. Moreover, the open box

$$U = \left\{ \frac{31}{64} < y < \frac{33}{64}, \quad \frac{15}{16} < u_j, v_j, a_j, b_j, c_j, d_j < 1 \right\}. \quad (48)$$

is captured by $\mathcal{M}$ in finite time. Indeed, in the envelope $7/16 < y < 9/16$, with the four core factors at least $15/16$ and the eight leaf factors above $7/8$, each unsaturated core factor has speed at least $13/256$ and reaches one before time $16/13$. During that interval $|\dot{y}| \leq 31/1024$ and every leaf speed has magnitude at most $1/64$; the total drifts are therefore less than $31/832$ and $1/52$, respectively, so the envelope is invariant until capture. Once the four core factors equal one, all projected speeds vanish. The normalized initial mass of (48) is

$$\mathbb{P}(U) = \frac{1}{32} \left(\frac{1}{16} \right)^{12} = 2^{-53}. \quad (49)$$

After Boolean rounding, exactly one of the two core clauses is violated, including under either tie convention for $y = 1/2$.

The unsigned motif has automorphism group of order 128, and its gauge orbit contains 2^{13} signed patterns. If X_{M6} counts isolated components in this orbit and $Q_N = 8 \binom{N}{3}$, then

$$\mathbb{E} X_{M6} = \frac{(N)_{13}}{128} 2^{13} \frac{\binom{8 \binom{N-13}{3}}{\binom{Q_N}{M}}}{\binom{Q_N}{M}}, \quad \frac{\mathbb{E} X_{M6}}{N} \longrightarrow \frac{729}{64} \alpha^6 e^{-39\alpha}. \quad (50)$$

For the independent-clause model, with $Q_{0,N} = 8 \binom{N-13}{3}$, the exact finite expectation is

$$\mathbb{E}X_{M6}^{\text{iid}} = \frac{(N)_{13}}{128} 2^{13} \frac{(M)_6 Q_{0,N}^{M-6}}{Q_N^M}, \quad \frac{\mathbb{E}X_{M6}^{\text{iid}}}{N} \rightarrow \frac{729}{64} \alpha^6 e^{-39\alpha}. \quad (51)$$

Component decoupling, the basin mass 2^{-53} , non-negativity of all other violations, and division by $M \sim \alpha N$ give (42). The same local calculation applies in the independent-clause model.

For the uniform model without replacement, changing one selected clause by one swap changes X_{M6} by at most 4: deleting a 3-clause can destroy at most one isolated M6 or expose at most three, while adding a 3-clause can destroy at most three isolated M6 components or create at most one. In the Doob exposure martingale, a transposition coupling therefore places the conditional expectations at each step in an interval of length at most 4. Hoeffding's martingale bound yields, with $d(\alpha) = (729/64)\alpha^6 e^{-39\alpha}$,

$$\mathbb{P}_F \left[X_{M6} < \frac{3}{4} d(\alpha) N \right] \leq \exp \left(-\frac{d(\alpha)^2}{512\alpha} N \right)$$

for all sufficiently large N . Conditional on such a formula, the certified M6 basin events are independent because isolated components have disjoint variable supports, and hence $Z_N \mid F \sim \text{Binomial}(X_{M6}, 2^{-53})$. A multiplicative Chernoff bound gives

$$\mathbb{P}_{x_0} \left[Z_N < \frac{3}{4} 2^{-53} X_{M6} \mid F \right] \leq \exp \left(-\frac{3 \cdot 2^{-53} d(\alpha)}{128} N \right).$$

Each captured component contributes at least one distinct violated clause, which proves (43) and the stronger conditional statement (44). $\square$

Remark 10.8 (Pointwise PDS convergence survives the refutation). *Let $X = [-1, 1]^N$ and $F = \Phi_{\text{mult}} + \delta_X$. The hypotheses of the nonsmooth KL dynamical theorem can be checked explicitly.*

First, F is proper, lower semicontinuous and semialgebraic, with nonempty domain X , and $F|_X = \Phi_{\text{mult}}|_X$ is continuous. Since δ_X is convex and subdifferentially regular and Φ_{mult} is C^1 , smooth-sum calculus gives

$$\hat{\partial}F(x) = \partial F(x) = \nabla \Phi_{\text{mult}}(x) + N_X(x).$$

The nonsmooth Kurdyka–Łojasiewicz inequality for lower-semicontinuous definable functions [2, Theorem 11] therefore supplies the KL property required in [1, Remark 4.8].

Second, Moreau decomposition gives, for $d(x) = \Pi_{T_X(x)}(-\nabla \Phi_{\text{mult}}(x))$,

$$-\dot{x} \in \partial F(x), \quad \|\dot{x}\| = \text{dist}(0, \partial F(x)),$$

and along every absolutely continuous solution

$$\frac{d}{dt} \Phi_{\text{mult}}(x(t)) = -\|\dot{x}(t)\|^2 \quad \text{a.e.}$$

The vector field $-\nabla \Phi_{\text{mult}}$ is Lipschitz on the compact box. For a nonempty closed convex constraint set, a projected dynamical system with Lipschitz vector field has a unique global absolutely continuous solution; equivalently, the normal-cone inclusion is a Lipschitz perturbation of a maximal monotone operator. Because trajectories remain in compact X , they are bounded, and $F \circ x = \Phi_{\text{mult}} \circ x$ is absolutely continuous.

Thus all requirements listed in [1, Remark 4.8] hold. The finite-length theorem [1, Theorem 4.5] yields

$$\int_0^\infty \|\dot{x}(t)\| dt < \infty,$$

and hence convergence to a single projected critical point. The rate statement is then covered by [1, Theorem 4.7]. This result does not imply zero limit energy: the open set (48) converges to $\mathcal{M} \subset \{\Phi_{\text{mult}} = 1\}$.

11 The Central Conjecture of the CLG-R Program

Conjecture 11.1 (Remaining CLG-R Comparison in the Clustering Regime). *For the uniform distinct-clause ensemble $\mathcal{E}_{\text{unif}}(N, \alpha)$ in the non-trivial clustering interval $\alpha \in (\alpha_d, \alpha_s)$, where $\alpha_d \approx 3.86$ is the clustering threshold [15] and $\alpha_s \approx 4.267$ is the satisfiability threshold, under pure projected gradient flow with uniform initialization, the remaining conjectural comparison is*

$$\liminf_{N \rightarrow \infty} \bar{\rho}_{N, \text{quad}}(\alpha) > \liminf_{N \rightarrow \infty} \bar{\rho}_{N, \text{mult}}(\alpha). \quad (52)$$

The positivity of the multilinear term is no longer conjectural in this ensemble: the M6 theorem gives

$$\liminf_{N \rightarrow \infty} \bar{\rho}_{N, \text{mult}}(\alpha) \geq \frac{729}{2^{59}} \alpha^5 e^{-39\alpha} > 0.$$

For planted ensembles, no corresponding M6 frequency theorem is asserted here; their residual comparison remains a separate open problem.

12 Summary Synthesis and Formal Status Matrix

Table 1: Consolidated CLG-R Mathematical Synthesis (Version 4.0.5 Audited).

Property	Quadratic Hinge (Φ_{quad})	Multilinear (Φ_{mult})	Softplus (Φ_{soft})
Regularity	$\mathcal{C}^1$ (piecewise quadratic)	$\mathcal{C}^\infty$ (polynomial)	$\mathcal{C}^\omega$ (real analytic)
Critical Measure $\mu(\mathcal{C}_0)$	> 0 (Plateau $\geq (1/3)^N$)	$= 0$ (Fubini / Polynomials)	$= 0$ (Real Analytic Identity)
Laplacian $\Delta\Phi$	$\equiv 0$ on $\mathcal{U}_N$	$\equiv 0$ on $\mathbb{R}^N$ (Harmonic)	$\text{Tr}(V^T W V) > 0$ (Subharmonic)
Attractor Loci	LP Polytope Z	Isolated stable points are vertices; positive boundary continua can attract	Unique minimizer if $\text{rank}(V) = N$
Physical Nature	Canonical LP Relaxation	Naive Mean-Field / p -spin	Smooth Hinge / log-sum-exp smoothing
Hessian Condition	Degenerate on $\mathcal{U}_N$	Zero diagonal	If $\text{rank}(V) = N$: $\kappa(H) \leq \kappa(W)\kappa(V^T V)$
Dynamic Trajectory	Universal centripetal flow	Projected Lyapunov decay	Strictly convex funnel if $\text{rank}(V) = N$
Dynamical Separation	Positive subcritical isolated-clause bound (Lemma 10.4, $0 < \alpha < 1/6$)	Former subcritical claim refuted in expectation and w.h.p.; M6 minimal on linear hypertrees; Horn DAGs open	Finite- β regularization

13 Epistemological Firewall: 3-XOR-SAT and the P versus NP Distinction

The CLG-R framework establishes an epistemological firewall against illegitimate inferences from continuous dynamical hardness to the P versus NP distinction:

- **3-XOR-SAT** is solvable in deterministic polynomial time $\mathcal{O}(N^3)$ via Gaussian Elimination over $\mathbb{F}_2$ (strictly in Class **P**).
- In the CLG computational experiments, the corresponding continuous multilinear dynamics exhibits severe glassy trapping despite the polynomial-time discrete solution method; related diluted p -spin/XORSAT literature provides the statistical-physics context for such metastable structure [19, 25].

Table 2: **Formal Status Matrix of CLG-R Theoretical Results (Audited Standard — Version 4.0.5).**

Result	Formal Audit Status
Theorem 1 (Central Box $\mathcal{U}_N$)	Closed (Deterministic, slack 0.5)
Theorem 2 (Measure Zero Critical Sets)	Closed under stated hypotheses (Fubini / Real Analytic)
Theorem 3 (Harmonicity and Saddles)	Closed with hypotheses (Harmonic, Milnor Curve Selection)
Theorem 4A' (Face Minima without H4)	Closed (Inherited vertex energies)
Corollary 4B (LaSalle Vertex Confinement)	Closed (isolated asymptotically stable equilibria are vertices)
Theorem 5 (Hessian Factorization)	Closed under stated conditions ($V^T W V \succeq 0$)
Theorem 6 (Lipschitz and IEEE 754)	Closed ($L_\beta = \Theta(\beta)$; idealized normal/subnormal/round-to-zero scales distinguished)
Theorem 7B (Centripetal Contraction)	Closed as Limit Set ($\mathcal{E}_{\text{proj}} \equiv Z$, $\lim_{t \rightarrow \infty} \text{dist}(x(t), Z) = 0$)
Proposition 7A (Negative Clause Families)	Closed for the Jacobian-sign claim ($J_{ij} \leq 0$ for purely negative clauses; global dynamics suspended)
Theorem 8 (Jensen Volume Bound)	Closed as finite-dimensional lower bound ($\mathbb{E}[\mu_{\text{norm}}(Z)] \geq (5/6)^{\lfloor \alpha N \rfloor} > 0$)
Theorem 9 (Horn Dynamics)	Open (former unit-fact trapping claim falsified; general Horn DAG dynamics unresolved)
Lemma 10.1 (Component Decoupling & Defect Tightness)	Partial (decoupling and degree identity remain; universal H_{leaf} is refuted)
Lemma 10.2 (Strict Saddle Subcritical)	Refuted as a universal claim (M6 has positive non-strict relative minima)
Lemma 10.3 (Degenerate Edge Non-Equilibrium)	Insufficient (a one-dimensional local statement cannot exclude higher-dimensional attracting faces)
Lemma 10.4 (Hinge Subcritical Basin Bound)	Closed (Exact $p_{\text{fail}} = 3/32$, $\rho_{\text{quad}} \geq \frac{3}{32} e^{-9\alpha} > 0$, Chebyshev concentration)
Theorem 10 (Subcritical 3-SAT Separation)	Refuted (positive M6 residual in expectation and w.h.p.; $m_{*,HT}^{\text{mult}} = 6$)
Central Conjecture	Reduced Comparison Conjecture (random ensemble: only $\rho_{\text{quad}} > \rho_{\text{mult}}$ remains open; planted case separate)
Epistemological Firewall (3-XOR-SAT)	Epistemological Result (Decoupling dynamic hardness from the P versus NP distinction)

- Therefore, failure or success of a continuous gradient flow neither implies nor is implied by Turing solvability.

14 Conclusion

We have established rigorous analytical foundations for Computational Landscape Geometry and Representation (CLG-R), including the Universal Centripetal Contraction Theorem, Exact Hessian Factorization, harmonic exclusion of interior minima, vertex confinement for isolated asymptotically stable multilinear equilibria, and a finite-dimensional LP-polytope volume bound. The adversarial M6 analysis changes the subcritical picture decisively: six clauses are necessary and sufficient for a positive-measure bad basin within connected linear 3-uniform Berge-acyclic components, and isolated M6 components occur with linear density in the audited sparse random ensemble. Consequently the multilinear residual is bounded away from zero both in expectation and with asymptotically high probability, refuting the former complete subcritical-separation theorem. The remaining random-ensemble conjecture is comparative rather than existential: whether the Hinge residual remains strictly larger than the multilinear residual in the clustering regime. None of these dynamical results implies a separation of computational complexity classes.

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